

Grade 7
Unit 8
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

In the table an expression has been evaluated, but the steps are mixed up. Order the expression using the numbers 1-5 on the left, the final answer has been given to you. Write an explanation for each step.

Step	Expression	Explain the Step
5	$150 - (17.64)$	Exponent
1	$ -9.392 + (-5.2)^3 - (-4.2)^2$	Original expression
4	$150 - (-4.2)^2$	Parenthesis
2	$ -9.392 + (-140.608) - (-4.2)^2$	Exponent within parenthesis
3	$ -150 - (-4.2)^2$	Simplify within parenthesis
Answer	132.36	Subtract

Evaluate each expression without using a calculator. Show work to support your answer.

6. $\left(\frac{4}{5}\right)^2 \times 25 + 4^2$

$$\begin{aligned} &\frac{16}{25} \times 25 + 4^2 \\ &\frac{16}{25} \times 25 + 16 \\ &16 + 16 \\ &32 \end{aligned}$$

$$7. \quad [(11 - 3.5)^2 - 6.25]^2 \div 50$$

$$\begin{aligned} & [7.5^2 - 6.25]^2 \div 50 \\ & [56.25 - 6.25]^2 \div 50 \\ & 50^2 \div 50 \\ & 2500 \div 50 \\ & 50 \end{aligned}$$

$$8. \quad 12.2 - (3^3 - 2)^2 \div 25$$

$$\begin{aligned} & 12.2 - (27 - 2)^2 \div 25 \\ & 12.2 - 25^2 \div 25 \\ & 12.2 - 625 \div 25 \\ & 12.2 - 25 \\ & -12.8 \end{aligned}$$

$$9. \quad -\left|4 - \frac{3}{4}\right| + \frac{3}{10}(3^2 - 49)$$

$$\begin{aligned} & -\left|3\frac{1}{4}\right| + \frac{3}{10}(9 - 49) \\ & -3\frac{1}{4} + \frac{3}{10}(-40) \\ & -3\frac{1}{4} - 12 \\ & -15\frac{1}{4} \end{aligned}$$

$$10. \quad \frac{(-8 + |-5.5 - 6.5|)^3}{16}$$

$$\frac{(-8 + |-12|)^3}{16}$$

$$\begin{aligned} & \frac{(-8 + 12)^3}{16} \\ & \frac{4^3}{16} \\ & \frac{64}{16} \\ & 4 \end{aligned}$$